# Practice Set 4 Paper 2 – Urology

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*Practice paper compiled from the question bank.*

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# Q1. Discuss treatment of 25 yr female with bladder TCC T1NoMo and C13

**Type:** LAQ (25 marks) (T1G3 management also answered in 2019-Summer Paper 2, Q1)

*"C13" is read as CIS (carcinoma in situ): T1 N0 M0 urothelial carcinoma with concurrent CIS in a 25-year-old woman.*

## Introduction and risk category

- TCC (transitional cell carcinoma), now urothelial carcinoma; T1 invades lamina propria; CIS is flat and always high grade.
- **T1 with CIS is high-risk NMIBC (non-muscle-invasive bladder cancer)** (AUA [American Urological Association] and EAU [European Association of Urology]). CIS is the second strongest progression factor after grade. Untreated CIS progresses to muscle invasion in 40–83%.
- **EAU 2021 very high risk:** T1 high grade (HG) with CIS plus at least one of age >70, multiple papillary tumours, or diameter ≥3 cm. In this woman, age does not count, so number and size decide. LVI (lymphovascular invasion) and variant histology also make it very high risk.
- **Young woman:** long life expectancy, fertility, pregnancy and sexual function shape every choice.

## Evaluation

- Smoking and occupational history; cytology (usually positive in CIS); CT (computed tomography) urography (upper-tract tumour <5%, more with CIS); examination under anaesthesia including vagina.
- **Pregnancy test** before CT and BCG (bacille Calmette-Guérin); expert pathology review.

## Step 1: Complete TURBT (transurethral resection of bladder tumour)

- Resect all tumour **with detrusor muscle**. **Blue-light cystoscopy** finds about one-third more CIS. Biopsy red velvety areas and the **bladder neck** (relevant to a later neobladder).

## Step 2: Re-TURBT within 6 weeks (all T1)

- Residual tumour in ≥40%; upstaging to muscle invasion 25–30% (half if no muscle). **Residual T1: progression up to 80%.**

```
T1 + CIS → re-TURBT within 6 weeks
  ├─ ≥T2 → neoadjuvant chemotherapy + cystectomy
  ├─ residual T1 / very high risk → offer early radical cystectomy
  └─ T0/Ta/CIS → BCG induction + 3-year maintenance
```

## Step 3a: BCG (bladder preservation; standard for most)

- **Mechanism:** BCG binds **fibronectin** and triggers a **Th1 (T-helper 1)** cytokine response (IFN-γ [interferon-gamma], IL-2 [interleukin-2]). It is the **only intravesical agent that reduces progression**.
- **Schedule:** start 2–4 weeks after TURBT, never immediately (intravasation and sepsis). Induction weekly ×6. **SWOG (Southwest Oncology Group) maintenance:** 3 weekly doses at 3 and 6 months, then 6-monthly to **3 years** (median recurrence-free survival 76.8 vs 35.7 months).
- **CIS:** tumour-free up to 84%, about half durable; persistent CIS at 3 months: second course gives a further 30–50% complete response.
- BCG is live: avoid pregnancy and breastfeeding during treatment.

- **Contraindications:** gross haematuria, traumatic catheterisation, symptomatic UTI (urinary tract infection). BCG sepsis: antituberculous drugs and steroids.

## Step 3b: Early (immediate) radical cystectomy

- **Rationale:** clinical T1 is understaged in one-third to one-half. Immediate cystectomy gives about **92% 10-year disease-free survival**. Patients who progress on surveillance do worse than those presenting with muscle invasion.
- **Offer for:** very high risk, residual HG T1, LVI, variants, diffuse CIS, or patient preference (AUA: fit patients with T1 and CIS).

## Step 3c: Cystectomy tailored to a young woman

- Standard female operation is anterior exenteration (bladder, urethra, anterior vagina, uterus, cervix, ovaries).
- **Organ-sparing** (≤cT2, bladder neck uninvolved): keep the **uterus and ovaries** (fertility, hormones, avoids surgical menopause; ovaries also make testosterone). Keep the **vagina and neurovascular tissue** (vaginal dissection no more dorsal than 2 and 10 o'clock) for sexual function and continence. Oncological safety of sparing is not fully known.
- **Orthotopic neobladder** suits her: needs a negative bladder-neck biopsy and **urethral frozen section** (negative bladder neck means negative urethra; CIS does not predict urethral involvement). Day continence 80–90%; **retention 20–30% in women**, so she must accept CIC (clean intermittent self-catheterisation). See Practice Set 4 Paper 2, Q3e.
- **Fertility:** if uterus or ovaries must go, offer oocyte or embryo cryopreservation first. Pregnancy after a neobladder is possible; caesarean uses a high midline incision to avoid the pouch.

## Step 4: BCG failure

- **BCG-unresponsive** (persistent or recurrent HG disease after adequate BCG, HG T1 at 3 months, or HG relapse within 6 months [Ta/T1] or 12 months [CIS]): **timely radical cystectomy**.
- If declined: **pembrolizumab** (KEYNOTE-057: CIS complete response 41%), **nadofaragene firadenovec** (53%), gemcitabine–docetaxel.

## Follow-up

- Bladder preserved: cystoscopy and cytology 3-monthly for 2 years, then 6-monthly, then yearly; regular upper-tract imaging. After neobladder: B12, metabolic panel, upper tracts.

**Also know:** EORTC (European Organisation for Research and Treatment of Cancer) and CUETO risk tables; TAR-200; checkpoint inhibitors with BCG (trials).

**Diagram to draw:** bladder wall layers showing CIS, Ta, T1 and T2.

> **Key points to remember:**
> - T1 + CIS = high risk; very high risk if multiple or ≥3 cm (EAU 2021).
> - Re-TURBT within 6 weeks; residual T1 means progression up to 80%.
> - BCG with 3-year maintenance, or early cystectomy (92% 10-year disease-free survival).
> - Young woman: organ-sparing cystectomy, neobladder, fertility preservation.

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# Q2. Anat. of Renal Vasculature + Dx modalities & Rx optns for Renovasc. HTN.

**Type:** LAQ (25 marks) (also answered in 2022-Summer Paper 2, Q2 and Practice Set 2 Paper 2, Q1)

## Part A: Anatomy of the renal vasculature

**Renal arteries**

- Arise from the aorta at the **L1–L2 disc**. The longer **right artery passes behind the IVC (inferior vena cava)**. Branches supply the adrenal, renal pelvis and proximal ureter.
- **Hilum, anterior to posterior:** renal vein, renal artery, pelvis; the posterior segmental artery lies behind the pelvis.
- **Anterior division** gives four segmental branches (apical, upper, middle, lower). **Posterior division** gives the posterior segmental artery: the first, most constant branch, passing behind the pelvis. If it crosses in front of the ureter, it can cause UPJO (ureteropelvic junction obstruction).
- **Five segmental arteries are end arteries** with no significant anastomosis. Occlusion causes segmental infarction; each segment can be resected separately.
- **Sequence:** segmental → lobar → interlobar (columns of Bertin) → arcuate (corticomedullary junction) → interlobular → afferent arteriole → glomerulus → efferent arteriole → peritubular capillaries or vasa recta.
- **Brödel line:** avascular plane between anterior and posterior territories, just behind the lateral border. Used for anatrophic nephrolithotomy and posterior calyceal access.
- **Variations (25–40%):** accessory arteries in 25–28% (T11–L4, rarely iliac or SMA [superior mesenteric artery]). They are end arteries, usually to the lower pole. Relevant to donor nephrectomy and endopyelotomy.

**Renal veins**

- Veins **anastomose freely** (venous collars, subcapsular plexus), so segmental vein ligation is safe. 3–5 segmental veins form the renal vein, anterior to the artery.

| | Right renal vein | Left renal vein |
|---|---|---|
| Length | 2–4 cm | 6–10 cm |
| Tributaries | None | Left adrenal, left gonadal, lumbar |
| Course | Direct to IVC | Between SMA and aorta (nutcracker syndrome) |
| Anomalies | Accessory veins commoner | Circumaortic (2–16%); retroaortic |

**Relevance to RVH (renovascular hypertension):** atheroma affects the **ostium and proximal third**; **FMD (fibromuscular dysplasia)** affects the **mid-distal artery and branches**; a stenosed accessory artery can cause segmental RVH.

## Part B: Diagnosis

**Definition:** RVH is hypertension caused by RAS (renal artery stenosis) and relieved by revascularisation. **Mechanism:** low perfusion pressure releases juxtaglomerular renin → angiotensin II (vasoconstriction) → aldosterone (sodium retention). Two-kidney one-clip is renin-dependent; one-kidney one-clip (solitary kidney or bilateral RAS) is volume-dependent.

**Clinical clues:** onset <30 years (FMD) or severe hypertension after 55; resistant hypertension (≥3 drugs); **creatinine rise >30% after ACE (angiotensin-converting enzyme) inhibitor or ARB (angiotensin receptor blocker)**; **flash pulmonary oedema**; kidneys differing by >1.5 cm; abdominal bruit; hypokalaemia.

| Test | Method and finding | Comment |
|---|---|---|
| **Duplex Doppler US (ultrasound)** | High peak systolic velocity; renal/aortic ratio >3.5; **RI (resistive index) ≥0.8 predicts poor response** | First line, no contrast; operator dependent |
| **CT angiography** | Stenosis, ostial plaque, beads | Best non-invasive anatomy; contrast nephropathy |
| **MR (magnetic resonance) angiography** | Gadolinium-enhanced | Caution if eGFR (estimated glomerular filtration rate) <30 mL/min |
| **Captopril renography (DTPA, diethylenetriaminepentaacetic acid)** | Day 1 after captopril: slow uptake, low peak; repeat off ACE inhibitor; ≥10% improvement = high probability | Captopril removes efferent constriction; declining use |
| **Catheter angiography** | Gold standard; **translesional gradient** | Done when intervention is planned |
| Renal vein renin | Ratio >1.5 lateralises | Before nephrectomy |

## Part C: Treatment

### 1. Medical therapy (first line for atherosclerotic RAS)

- ACE inhibitor or ARB (check creatinine and potassium in 1–2 weeks), calcium channel blockers, diuretics.
- **Statin, antiplatelet, smoking cessation, diabetes control**: most deaths are cardiovascular.

### 2. Landmark trials

- **ASTRAL (2009)** and **CORAL (2014):** stenting added **no benefit** over medical therapy for renal function, cardiovascular events or death. Hence no routine revascularisation of atherosclerotic RAS.

**3. Indications for revascularisation:** FMD with hypertension; atherosclerotic RAS with **flash pulmonary oedema**, **rapidly declining function** (bilateral or solitary-kidney RAS) or truly resistant hypertension; kidney must be viable (RI <0.8, not atrophic).

### 4. Endovascular

- **PTRA (percutaneous transluminal renal angioplasty) alone for FMD**; stent only for dissection or failure.
- **Ostial atheroma: angioplasty + stent**, because a balloon alone fails from elastic recoil of aortic plaque. Restenosis is the main late problem.
- Complications: contrast nephropathy, atheroembolism, dissection, thrombosis.

**5. Surgery** (failed endovascular, branch disease, aneurysm, children, aortic surgery)

| Technique | Use |
|---|---|
| Aortorenal bypass (saphenous vein; internal iliac artery in children) | Standard |
| Transaortic endarterectomy | Bilateral ostial plaque |
| Splenorenal (left) or hepatorenal (right) bypass | Diseased aorta |
| Ex vivo bench repair + autotransplantation | Branch FMD, aneurysm |
| Nephrectomy | Small non-functioning kidney with lateralising renin |

**Follow-up:** blood pressure, creatinine, duplex for restenosis.

**Diagram to draw:** renal artery dividing into anterior (four segments) and posterior divisions, with the Brödel line.

**Key points to remember:**

- Five segmental end arteries; posterior segmental passes behind the pelvis; veins anastomose freely.
- Left renal vein 6–10 cm, receives adrenal, gonadal and lumbar veins.
- Duplex first; angiography with gradient before intervention.
- ASTRAL and CORAL: no routine stenting; FMD gets balloon angioplasty without stent.

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# Q3a. Renal transplant in neurogenic bladder

**Type:** SAQ (10 marks)

## Why the bladder matters

- A neurogenic bladder with **poor compliance, low capacity or detrusor overactivity** transmits high pressure to the graft even without reflux. Poor emptying adds infection. Recurrent pyelonephritis, not augmentation, causes graft loss.
- **Five goals:** unimpeded drainage, low-pressure capacious storage, effective emptying, no infection, fewest procedures.

## Pretransplant evaluation

- Renal ultrasound, VCUG (voiding cystourethrogram) and **urodynamics** (capacity, compliance, overactivity, emptying, sphincter). Repeat them; the bladder changes over time.
- **Target:** storage to expected capacity at detrusor pressure **<40 cm $H_2O$**.
- **Defunctionalised bladder** (anuria or diversion): **bladder cycling** (saline fills of increasing volume via catheter), then urodynamics.

## Pretransplant optimisation (stepwise)

```
Antimuscarinic (oxybutynin), mirabegron, botulinum toxin
  → CIC (clean intermittent catheterisation) ± overnight drainage
  → bladder cycling if defunctionalised
  → still >40 cm H₂O, <75% expected capacity or CIC ≤3-hourly → augmentation
```

- **Augmentation:** ileum or sigmoid most used; dilated ureter (ureterocystoplasty) preferred when available (no mucus or metabolic effects). **Stomach avoided** (haematuria–dysuria syndrome in anuric patients). Augmentation does not clearly raise transplant complications.
- **Mitrofanoff** channel if urethral CIC is difficult; site it **medial** to the planned transplant incision.
- Do reconstruction **before** transplant; allow **6–12 weeks** to heal. Neuropathic bladders may deteriorate under high graft output, so earlier reconstruction is justified.
- **Native nephrectomy** only for named reasons: recurrent upper-tract infection, massive reflux, stones, uncontrolled hypertension.

## At transplant

- Extravesical non-refluxing ureteroneocystostomy into bladder or augment (know the augment pedicle). Ureteroureterostomy to a native ureter is an option.
- Ileal conduit drainage if the bladder is unusable.

## Post-transplant care

- Continue CIC; colonisation is expected, so treat only symptomatic UTI (urinary tract infection).
- **Creatinine rise with hydronephrosis:** catheter drainage for 1–2 weeks first. If it improves, the bladder is the cause; otherwise stent or nephrostomy and consider rejection.
- Febrile UTI with graft reflux: reimplant after optimising the bladder.
- Lifelong surveillance of augments (malignancy, stones).

**Diagram to draw:** stepwise bladder optimisation ladder.

**Key points to remember:**

- Target <40 cm $H_2O$ storage before transplant.
- Drugs → CIC → cycling → augmentation; reconstruct 6–12 weeks before.
- Pyelonephritis, not augmentation, loses grafts.

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## Q3b. Urolithiasis in pregnancy

**Type:** SAQ (10 marks) (also answered in 2022-Summer Paper 2, Q3d)

### Background

- 1 in 200 to 1 in 1500 pregnancies; **commonest non-obstetric cause of admission**; mostly second or third trimester.
- Linked with preterm delivery, pre-eclampsia and premature rupture of membranes.
- **Calcium phosphate stones:** raised GFR (glomerular filtration rate) and placental calcitriol cause hypercalciuria; urine is alkaline. Raised citrate, magnesium and urine flow protect, so stone rates barely change.
- **Physiological hydronephrosis** (up to 90% by third trimester, right-sided) mimics obstruction.

### Evaluation

- Urinalysis, **urine culture always**, blood count, creatinine; obstetric review.
- **US (ultrasound) first line**; transvaginal US for distal ureter; a rise in Doppler **RI (resistive index)** suggests acute obstruction.
- Second line: **MRI (magnetic resonance imaging)** (stone as a filling defect) or **low-dose CT (computed tomography)**. Keep total fetal dose **≤50 mGy**.

### Management

```
Symptomatic stone in pregnancy
 ├─ sepsis, solitary kidney, renal failure, uncontrolled pain
 │     → urgent stent or nephrostomy + antibiotics
 ├─ pain controlled, no infection → observe (50–80% pass)
 └─ failed observation → ureteroscopy + holmium laser
        or temporary stent/nephrostomy until delivery
```

- **Conservative:** hydration, paracetamol, small doses of morphine. **NSAIDs (non-steroidal anti-inflammatory drugs) contraindicated** (ductus closure in third trimester). **MET (medical expulsive therapy)** is off-label.
- **URS (ureteroscopy):** first-line intervention for non-complex stones in any trimester. Neuraxial anaesthesia, minimal fluoroscopy (lead shield beneath the patient, US guidance). Third-trimester URS may raise preterm labour.
- **Stent or PCN (percutaneous nephrostomy):** temporising. Rapid encrustation: change stents about **every 4 weeks**, nephrostomy about every 3 weeks; definitive treatment after delivery.
- Asymptomatic renal stones: observe until after delivery.
- **SWL (shock wave lithotripsy) contraindicated. PCNL (percutaneous nephrolithotomy) deferred** except in expert centres.

### After delivery

- Clear residual stones; **metabolic evaluation**.

**Diagram to draw:** the management flowchart above.

**Key points to remember:**

- US first; MRI or low-dose CT second (≤50 mGy).
- Observation first; no NSAIDs.
- URS with laser is definitive; SWL contraindicated.
- Stents encrust: change every 4 weeks.

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## Q3c. MIBG scan

**Type:** SAQ (10 marks)

### Principle

- MIBG (meta-iodobenzylguanidine) is a guanethidine analogue that resembles **noradrenaline**.
- It enters chromaffin and sympathetic cells through the **noradrenaline transporter** and is stored in **adrenergic secretory vesicles**. So it maps catecholamine-handling tissue: function, not just anatomy.

### Isotopes

| | **$^{123}$I-MIBG** | **$^{131}$I-MIBG** |
|---|---|---|
| Use | **Diagnostic** (preferred) | Older diagnostic; mainly **therapeutic** |
| Image quality | Better; allows SPECT-CT (single-photon emission computed tomography) | Poorer |
| Emission | Gamma | Beta (therapy) plus gamma |

### Technique

- **Thyroid blockade** with potassium iodide before and after injection (free iodine).
- **Stop interfering drugs:** labetalol, tricyclic antidepressants, sympathomimetics, reserpine, some calcium channel blockers.
- Intravenous injection; planar and SPECT-CT images at about 24 hours.
- **Normal uptake:** salivary glands, liver, heart, bladder, thyroid if unblocked; normal adrenals faint.

### Uses

#### 1. Phaeochromocytoma and paraganglioma

- **Extra-adrenal, multiple, metastatic or recurrent disease**; biochemistry positive but imaging negative.
- **Isolated normetanephrine excess** may indicate an extra-adrenal tumour; MIBG finds it.
- Tumour >5 cm: MIBG or FDG (fluorodeoxyglucose) PET (positron emission tomography) to exclude metastases before surgery.
- **Best in MEN-2 (multiple endocrine neoplasia type 2).** Raised metanephrines after surgery: MIBG can unmask metastases.

#### 2. Neuroblastoma

- $^{123}$I-MIBG is **required by COG (Children's Oncology Group) for staging**, replacing skeletal survey and bone scan. More sensitive for relapse. Non-avid tumours: PET.

#### 3. Therapy ($^{131}$I-MIBG)

- Confirm uptake first. Metastatic phaeochromocytoma: symptom response in up to two-thirds; complete response <5%. High-specific-activity form FDA (Food and Drug Administration)-approved 2018.
- Relapsed or refractory neuroblastoma: response about 28–39%.

### Limitations

- **Cold scans in SDHB (succinate dehydrogenase subunit B) and VHL (von Hippel–Lindau)**; aggressive metastases lose uptake.

- **FDG PET is now gold standard** for staging, and **$^{68}$Ga-DOTATATE PET/CT** is replacing it at some centres.
- **Also know:** carcinoid and medullary thyroid cancer imaging; cardiac sympathetic imaging.

**Diagram to draw:** whole-body MIBG showing an adrenal hot spot with normal liver, heart and bladder uptake.

> **Key points to remember:**
> - Noradrenaline analogue; block thyroid; stop labetalol.
> - $^{123}$I diagnostic, $^{131}$I therapeutic.
> - Best for extra-adrenal, metastatic and MEN-2 tumours; cold in SDHB and VHL.

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## Q3d. Prostate biopsy

**Type:** SAQ (10 marks) (TRUS biopsy also answered in 2020-Summer Paper 2, Q3d)

### Indications

- Raised PSA (prostate-specific antigen) with risk factors; suspicious DRE (digital rectal examination); MRI (magnetic resonance imaging) PIRADS (Prostate Imaging Reporting and Data System) ≥3; suspected metastasis.
- **Repeat:** rising PSA; ASAP (atypical small acinar proliferation; 50% cancer risk); positive biomarkers (PHI [Prostate Health Index], 4Kscore, PCA3); active surveillance.
- **Contraindications:** coagulopathy, acute UTI (urinary tract infection) or prostatitis, severe immunosuppression. No anus: transperineal route.

### MRI first

- **PROMIS (2017):** MRI sensitivity 93% vs 48% for TRUS (transrectal ultrasound) biopsy. **PRECISION (2018):** targeted biopsy found more significant cancer (38% vs 26%) and avoided biopsy in 28%.
- Targeting: cognitive, MRI/US fusion or in-bore; 2–3 cores per target ± systematic cores.

### Preparation

- Continue low-dose aspirin; stop warfarin and clopidogrel 5–7 days, DOACs (direct oral anticoagulants) 2–5 days before.
- **Transrectal: antibiotics for all** (contaminated route), ≤24 hours: fluoroquinolone, or cephalosporin ± aminoglycoside. Augment if antibiotics in last 6 months, diabetes or previous infection.
- **Periprostatic block:** 1–2% lidocaine, 5 mL each side at the seminal vesicle–prostate junction.

### Technique

| | Transrectal | Transperineal |
|---|---|---|
| Position | Left lateral | Lithotomy |
| Needle route | Rectal wall (contaminated) | Sterile perineal skin |
| Antibiotics | Mandatory | Skin-flora only or none |
| Anterior/apex | Poor | Good |
| Anaesthesia | Periprostatic block | Perineal and periapical triangle block |

- 18-gauge spring-loaded needle. Indent the capsule, withdraw slightly and fire, so the core samples peripheral zone.
- **12 cores:** base, mid and apex on each side, medial and lateral; sextant missed lateral cancers. Saturation only after negative biopsies.
- **EAU (European Association of Urology) 2023 prefers transperineal biopsy.**

### Complications (transrectal)

| Complication | Rate |
|---|---|
| Haematuria | 23–63% |
| Haematospermia | 10–50% |
| Rectal bleeding | 2–22% |
| Infection | 0.1–7%; sepsis 0.3–3% |

| Complication | Rate |
|---|---|
| Retention | 0.2–0.4% |

- **Sepsis** (fluoroquinolone-resistant *E. coli*): carbapenem until cultures return. Rectal bleeding: pressure, balloon tamponade, endoscopy, embolisation.

**Also know:** PSA density; TZ (transition zone) cores if gland >50 mL; PSMA (prostate-specific membrane antigen) PET.

**Diagram to draw:** 12-core template on a transverse prostate outline.

> **Key points to remember:**
> - MRI before biopsy (PROMIS, PRECISION).
> - 12 cores; transrectal antibiotics ≤24 hours.
> - Bleeding commonest; sepsis most serious; transperineal preferred.

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## Q3e. Neobladder

**Type:** SAQ (10 marks) (also answered in 2020-Summer Paper 4, Q3a)

### Definition

- An **orthotopic** detubularised bowel (usually ileal) reservoir joined to the native urethra after radical cystectomy: **volitional voiding**, no stoma. 80–90% of men and about 75% of women are medically eligible.

### Principles

1. **Reliable outlet:** intact rhabdosphincter (pudendal nerve), healthy urethra.
2. **Low pressure:** complete **detubularisation** abolishes coordinated contractions; folding into a **sphere** gives most volume for surface area.
3. **Adequate volume:** <200 mL initially, **500–600 mL** by 1 year.

### Selection

- **Absolute contraindications:** tumour at the **urethral margin** (frozen section), urethral stricture, eGFR (estimated glomerular filtration rate) <35–40 mL/min, severe liver disease, inability to self-catheterise.
- **Relative:** prostatic stromal invasion (men); bladder-neck or anterior vaginal tumour (women); poor sphincter; irradiated pelvis; frailty. Age, obesity and CIS (carcinoma in situ) are not bars.

### Types

| Design | Segment | Ureters |
|---|---|---|
| **Studer** | 40–44 cm ileum + afferent limb | Refluxing, isoperistaltic limb |
| **Hautmann W** | About 70 cm ileum | Refluxing; largest capacity |
| T pouch | 44 cm + T limb | Antireflux flap valve |
| Camey II; Padua | Ileum | Le Duc antireflux |
| Mainz III; sigmoid | Ileocaecal; colon | Tunnelled |

### Studer technique (brief)

- Ileum taken 15–20 cm proximal to the ileocaecal valve. Open the distal 40–44 cm antimesenterically, sew into a U, fold into a sphere. Stented end-to-side ureteric anastomoses to the afferent limb; dependent point to urethra.

### Outcomes

- **Day continence 80–90%** (3–6 months); **night incontinence 20–50%** (no sphincter reflex, mucosal water secretion).
- **Retention:** men 4–10%, **women 20–30%**: men leak, women fail to empty.

### Complications

- **Early:** urine leak, ileus, infection, mucus plugging.
- **Late: hyperchloraemic metabolic acidosis**, **vitamin B12** deficiency, ureteroileal stricture (<3%), stones, pouch rupture, incisional hernia, urethral recurrence.

### Care and follow-up

- Mucus irrigation, bicarbonate; voiding training (sit, relax, strain), timed voiding, night alarm.

- Annual metabolic panel, B12, upper tracts; urethral cytology if CIS.

**Diagram to draw:** Studer U-pouch with afferent limb and urethral anastomosis.

> **Key points to remember:**
> - Detubularise and make a sphere; 500–600 mL by 1 year.
> - Absolute bar: positive urethral margin, stricture, poor renal function.
> - Day continence 80–90%, night 20–50%; women retain (20–30%).

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## Q3f. Role of immunotherapy in RCC

**Type:** SAQ (10 marks) (also answered in 2024-Winter Paper 2, Q3e and 2025-Winter Paper 4, Q3a)

### Rationale

- RCC (renal cell carcinoma) is immunogenic (rare spontaneous regression, T-cell infiltration) and chemoresistant.
- **Checkpoints:** PD-1 (programmed death-1) on T cells binds PD-L1 on tumour and switches T cells off; CTLA-4 (cytotoxic T-lymphocyte antigen-4) damps priming. Blockade **releases the brake** on existing immunity.
- VEGF (vascular endothelial growth factor) suppresses immunity, so TKIs (tyrosine kinase inhibitors) synergise.

### Cytokine era

- IFN-α (interferon-alpha): response 10–15%; obsolete.
- **High-dose IL-2 (interleukin-2):** durable complete response **7–9%**; capillary leak needs intensive care; selected fit patients only.

### Checkpoint inhibitors (clear cell)

| Trial | Regimen | Result |
|---|---|---|
| CheckMate 025 | Nivolumab vs everolimus, after TKI | OS (overall survival) 25.0 vs 19.6 months |
| **CheckMate 214** | Nivolumab + ipilimumab vs sunitinib | Intermediate/poor risk: OS HR (hazard ratio) 0.63; complete response 9% |
| KEYNOTE-426 | Pembrolizumab + axitinib | Better PFS (progression-free survival) and OS |
| CheckMate 9ER | Nivolumab + cabozantinib | PFS 16.6 vs 8.3 months |
| **CLEAR** | Pembrolizumab + lenvatinib | PFS 23.9 vs 9.2 months; response 71% |

- **First line:** IO/TKI (immuno-oncology plus TKI) for all IMDC (International Metastatic RCC Database Consortium) risk groups; nivolumab + ipilimumab for intermediate/poor risk only.
- **After IO:** switch to a VEGF TKI; IO rechallenge failed (CONTACT-03). Belzutifan after IO and TKI.

### Adjuvant

- **KEYNOTE-564:** pembrolizumab for 1 year after nephrectomy in high-risk clear cell RCC, including resected metastases; disease-free survival HR 0.63, later OS benefit.

### Toxicity: irAEs (immune-related adverse events)

- Colitis, hepatitis, pneumonitis, thyroiditis, hypophysitis, nephritis, myocarditis.

| Grade | Action |
|---|---|
| 1 | Continue, monitor |
| 2 | Hold; prednisolone 0.5–1 mg/kg |
| 3–4 | Stop; methylprednisolone 1–2 mg/kg; infliximab if refractory |

### Biomarkers

- **PD-L1 does not select patients.** Sarcomatoid tumours respond well.
- **Also know:** avelumab + axitinib (JAVELIN, no OS benefit); non-clear cell data scant.

**Diagram to draw:** T cell–tumour synapse with PD-1/PD-L1 and CTLA-4/B7 blocked.

**Key points to remember:**

- IO/TKI for all risks; nivolumab + ipilimumab for intermediate/poor only.
- CheckMate 214: OS HR 0.63, complete response 9%.
- KEYNOTE-564: adjuvant pembrolizumab.

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# Rapid Revision Sheet – Practice Set 4 Paper 2

*All the key points from this paper in one place. Read this in the last 30 minutes before the exam.*

**Q1. Discuss treatment of 25 yr female with bladder TCC T1NoMo and C13 (LAQ)**

- T1 + CIS = high risk; very high risk if multiple or ≥3 cm (EAU 2021).
- Re-TURBT within 6 weeks; residual T1 means progression up to 80%.
- BCG with 3-year maintenance, or early cystectomy (92% 10-year disease-free survival).
- Young woman: organ-sparing cystectomy, neobladder, fertility preservation.

**Q2. Anat. of Renal Vasculature + Dx modalities & Rx optns for Renovasc. HTN. (LAQ)**

- Five segmental end arteries; posterior segmental passes behind the pelvis; veins anastomose freely.
- Left renal vein 6–10 cm, receives adrenal, gonadal and lumbar veins.
- Duplex first; angiography with gradient before intervention.
- ASTRAL and CORAL: no routine stenting; FMD gets balloon angioplasty without stent.

**Q3a. Renal transplant in neurogenic bladder (SAQ)**

- Target <40 cm $H_2O$ storage before transplant.
- Drugs → CIC → cycling → augmentation; reconstruct 6–12 weeks before.
- Pyelonephritis, not augmentation, loses grafts.

**Q3b. Urolithiasis in pregnancy (SAQ)**

- US first; MRI or low-dose CT second (≤50 mGy).
- Observation first; no NSAIDs.
- URS with laser is definitive; SWL contraindicated.
- Stents encrust: change every 4 weeks.

**Q3c. MIBG scan (SAQ)**

- Noradrenaline analogue; block thyroid; stop labetalol.
- $^{123}I$ diagnostic, $^{131}I$ therapeutic.
- Best for extra-adrenal, metastatic and MEN-2 tumours; cold in SDHB and VHL.

**Q3d. Prostate biopsy (SAQ)**

- MRI before biopsy (PROMIS, PRECISION).
- 12 cores; transrectal antibiotics ≤24 hours.
- Bleeding commonest; sepsis most serious; transperineal preferred.

**Q3e. Neobladder (SAQ)**

- Detubularise and make a sphere; 500–600 mL by 1 year.
- Absolute bar: positive urethral margin, stricture, poor renal function.
- Day continence 80–90%, night 20–50%; women retain (20–30%).

**Q3f. Role of immunotherapy in RCC (SAQ)**

- IO/TKI for all risks; nivolumab + ipilimumab for intermediate/poor only.
- CheckMate 214: OS HR 0.63, complete response 9%.
- KEYNOTE-564: adjuvant pembrolizumab.